

1. 考虑如下弹簧模型

(a) 求弹簧伸缩

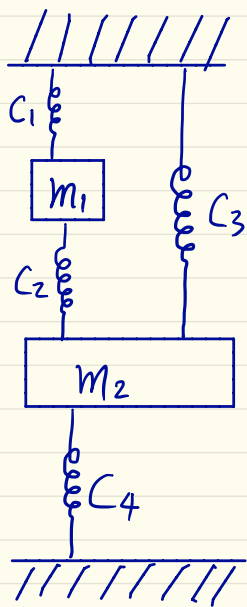
$e = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{pmatrix}$ 和质体位移

$u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ 的关系 $e = Au$.

(b) 求刚度矩阵

$$K = A^T C A.$$

(c) K 是正定阵吗?



2. 求解微分方程和差分方程组

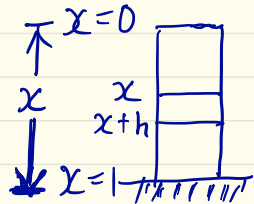
$$\textcircled{1} -\frac{d^2 u}{dx^2} = 1, \quad u'(0)=0, u(1)=0$$

$$\textcircled{2} -\frac{u_{i+1} + 2u_i - u_{i-1}}{h^2} = 1, \quad i=1, \dots, n$$

满足 $\frac{u_1 - u_0}{h} = 0, \quad u_{n+1} = 0.$
 $h = \frac{1}{n+1}$

注: ① 对应 free-fixed ends

模型, $c(x)=1, f(x)=1$



② 对应着 ① 的离散形式:

n 个质体通过 n 个弹簧相连

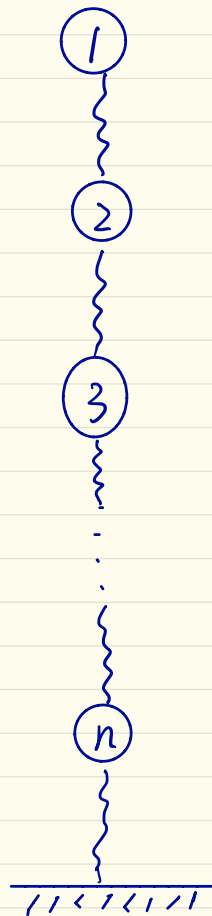
$$e_0 = u_1 - u_0 = 0$$

$$e_1 = u_2 - u_1$$

$$e_2 = u_3 - u_2$$

$\vdots$

$$e_n = u_{n+1} - u_n \quad u_{n+1} = 0$$



3. 考虑 n 个质体被 $n+1$ 个弹簧相连的 fixed-fixed ends 模型

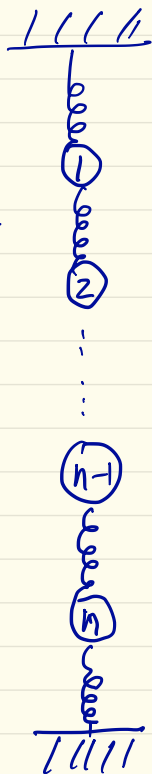
(1) 求弹簧伸缩向量

$e = \begin{pmatrix} e_1 \\ e_2 \\ \vdots \\ e_{n+1} \end{pmatrix}$ 和质体位移向量

$u = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$ 的关系 $e = Au$

(2) $K = A^T A$ 是正定阵.

(3) K^{-1} 是正矩阵.



4. 给定一个 n 阶阵 A , 回忆

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

设 $\rho(A)$ 是 A 的特征值长度
最大值.

证明:
$$\rho(A) = \lim_{k \rightarrow +\infty} \|A^k\|^{\frac{1}{k}}$$

(Gelfand 公式)

5. 设 A 是 n 阶非负矩阵, 且存在 $k \in \mathbb{N}$, A^k 是正矩阵,

证明: (1) $\rho(A)$ 是 A 的特征值且它的代数重数和几何重数均为 1. (注: $\rho(A) = \max\{|\lambda_i| \mid \lambda_i \text{ 是 } A \text{ 特征值}\}$)

且存在 $x \in \mathbb{R}^n$, $x > 0$,
 $Ax = \rho(A)x$.

(2) 若 $y \geq 0$, $y \neq 0$ 满足

$$Ay \leq cy, \quad c \in \mathbb{R}$$

则 $y > 0$, 且 $c \geq P(A)$.

6. 设 A 是 n 阶正矩阵, 即 $A > 0$,
 $\rho(A)$ 是谱半径, 即 $\rho(A) = \max\{|\lambda_i| \mid \lambda_i \text{ 是 } A \text{ 的特征值}\}$, 由 Perron-Frobenius
定理, 存在 $v = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} > 0$, $Av = \rho(A)v$
且 $(1, \dots, 1)v = 1$ (即 $a_1 + \dots + a_n = 1$)

证明: (1) 存在 $w = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} > 0$,

$$w^T A = \rho(A) w^T, \text{ 且 } w^T v = 1.$$

(2) 令 $B = \frac{1}{\rho(A)} \cdot A$, 则

$$\lim_{t \rightarrow +\infty} B^t = v w^T$$

(3) 设 $Au = \lambda_0 u$, 且 $|\lambda_0| < \rho(A)$.

则 $w^T u = 0$

(4) 给定 $x_0 \geq 0 \in \mathbb{R}^n$, 且 $x_0 \neq 0$,

令 $x_1 = Bx_0$, $x_2 = Bx_1$, $\dots$,

$x_{t+1} = Bx_t$, 则

$$\lim_{t \rightarrow +\infty} x_t = (w^T x_0) v$$

7. 设 $A = (a_{ij})_{n \times n}$, $B = (b_{ij})_{n \times n}$,

我们写 $B > A$, 若 $a_{ij} < b_{ij}$,

$\forall i, j = 1, \dots, n$. 现在假设

$B > A > 0$, 即 $b_{ij} > a_{ij} > 0$,

$\forall i, j = 1, \dots, n$. 证明:

$$\rho(B) > \rho(A)$$

这里 $\rho(A) = \max\{|\lambda| \mid \lambda \text{ 是 } A \text{ 的特征值}\}$.

8. (Leslie model)

设 $f(t) = \begin{pmatrix} f_1(t) \\ \vdots \\ f_n(t) \end{pmatrix}$ 是在时间 t 的人口向量, $f_i(t)$ 是年龄组 i 的人口数 (1-20岁, 21~40岁, 41~60岁等年龄组) 令

$$L = \begin{pmatrix} b_1 & b_2 & \cdots & b_n \\ s_1 & 0 & \cdots & 0 \\ & s_2 & & \vdots \\ & & \ddots & \vdots \\ & & & s_{n-1} & 0 \end{pmatrix}$$

其中 b_k 是 birth rate, s_k 是 survival

rate

$$\text{令 } f(t+1) = L f(t)$$

即在时间 $t+1$ 的人口向量.

$$\text{令 } \bar{F}_k(t) = \frac{f_k(t)}{f_1(t) + \cdots + f_n(t)}$$

即在时间 t , 年龄组 k 的人口占总人口的百分率.

$$\text{令 } \bar{F}(t) = (\bar{F}_1(t), \cdots, \bar{F}_n(t))^T$$

求 $\lim_{t \rightarrow +\infty} \bar{F}(t)$.

参 考 答 案

$$\begin{aligned}
 (a) \quad e_1 &= u_1 \\
 e_2 &= u_2 - u_1 \\
 e_3 &= u_2 \\
 e_4 &= -u_2
 \end{aligned}
 \Rightarrow e = \begin{pmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \\ 0 & -1 \end{pmatrix} u$$

$$(b) \quad A^T C A = K = \begin{pmatrix} c_1 + c_2 & -c_2 \\ -c_2 & c_2 + c_3 + c_4 \end{pmatrix}$$

$$(c) \quad x^T K x = x^T A^T C A x$$

$$\begin{aligned}
 \hat{=} y &= Ax & \text{上式} &= y^T C y \\
 &= \begin{pmatrix} y_1 \\ \vdots \\ y_4 \end{pmatrix} & &= c_1 y_1^2 + \dots + c_4 y_4^2
 \end{aligned}$$

$$\Rightarrow x^T K x \geq 0$$

若 $x^T K x = 0$ 即 $y^T C y = 0$

$\Rightarrow y = 0 \Rightarrow x = 0$ (因为 A 列

线性无关).

2.

$$\textcircled{1} \quad u(x) = -\frac{1}{2}x^2 + Cx + D$$

$$\text{代入 } u'(0) = 0 \Rightarrow C = 0$$

$$\text{代入 } u(1) = 0 \Rightarrow D = \frac{1}{2}$$

$$\Rightarrow u(x) = \frac{1}{2}(1 - x^2)$$

$$\textcircled{2} \quad i=1, \Rightarrow -u_2 + u_1 = 1$$

$\vdots$

$$i=n \Rightarrow 2u_n - u_{n-1} = 1$$

即

$$\frac{1}{h^2} \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & -2 & -1 \\ & & & -1 & 2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

$$\text{令 } K = \begin{pmatrix} 1 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & \ddots \\ & & & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}_{n \times n}$$

K 是可逆阵:

方法1. 行消去

方法2. $K = A^T A$

$$A = \begin{pmatrix} 1 & -1 & & & \\ 0 & 1 & -1 & & \\ & & \ddots & \ddots & \\ & & & \ddots & -1 \\ & & & & 0 & 1 \end{pmatrix}_{n \times n} \quad (\text{由习题的注可得 } A)$$

A 显然可逆

$$\Rightarrow u = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = h^2 K^{-1} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

$$= h^2 A^{-1} (A^T)^{-1} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

$$= h^2 \begin{pmatrix} 1 & 1 & \cdots & 1 \\ & 1 & \cdots & 1 \\ & & \ddots & \vdots \\ & & & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & \cdots & 1 \\ & 1 & \cdots & 1 \\ & & \ddots & \vdots \\ & & & 1 \end{pmatrix} \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}$$

$$= h^2 \begin{pmatrix} 1+2+\cdots+n \\ 2+\cdots+n \\ \vdots \\ n \end{pmatrix}$$

$$\Rightarrow u_i = \frac{1}{2} h^2 (n+i)(n+1-i)$$

比较

$$u''(x) \text{ 和 } \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

的误差

$$u(x+h) = u(x) + h u'(x) + \frac{1}{2} u''(x) h^2 + \frac{1}{6} h^3 u'''(x) + \dots$$

$$u(x-h) = u(x) - h u'(x) + \frac{1}{2} h^2 u''(x) - \frac{1}{6} h^3 u'''(x) + \dots$$

$$\Rightarrow \frac{u(x+h) - 2u(x) + u(x-h)}{h^2}$$

$$= u''(x) + \underbrace{ch^2 u^{(4)}(x) + \dots}_{\text{误差}}.$$

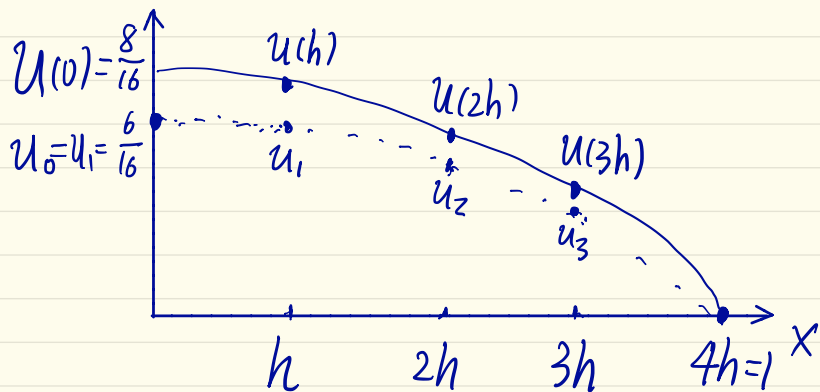
$$= u''(x) + O(h^2) :-$$

比较 $u'(x)$ 和 $\frac{u(x+h) - u(x)}{h}$.

$$\frac{u(x+h) - u(x)}{h} = u'(x) + O(h)$$

比较 u_i 和 $u(ih)$

$$\text{当 } n=3, \quad h = \frac{1}{4}$$



$$u(ih) = \frac{1}{2} [1 - (ih)^2] = \frac{1}{2} \left[1 - \frac{i^2}{(n+1)^2} \right]$$

$$= \frac{1}{2} \cdot \frac{(n+1+i)(n+1-i)}{(n+1)^2}$$

$$u_i = \frac{1}{2} h^2 (n+1+i)(n+1-i) = \frac{1}{2} \frac{(n+1-i)(n+1+i)}{(n+1)^2}$$

$$u(ih) - u_i = \frac{1}{2} \frac{(n+1-i)}{(n+1)^2}$$

$$= \frac{1}{2} h (1 - ih)$$

$$3. \quad (1) \quad e_1 = u_1 - u_0 \quad u_0 = 0$$

$$e_2 = u_2 - u_1$$

$$\vdots$$

$$e_n = u_n - u_{n-1}$$

$$e_{n+1} = -u_n$$

$$\Rightarrow e = \begin{pmatrix} 1 \\ -1 & 1 \\ & -1 & 1 \\ & & \ddots & \ddots \\ & & & -1 & 1 \\ & & & & -1 \end{pmatrix} u$$

$(n+1) \times n$

$$= Au$$

$$(2) \text{ 求解 } Ax = 0 \Rightarrow x = 0$$

$$\Rightarrow A \text{ 列满秩}$$

$$\hat{K} = A^T A$$

$$\forall x \neq 0, \quad x^T K x = x^T A^T A x$$

$$\text{令 } Ax = y = \begin{pmatrix} y_1 \\ \vdots \\ y_{n+1} \end{pmatrix} \neq 0 \quad (A \text{ 列满秩})$$

$$\Rightarrow x^T K x = y_1^2 + \dots + y_{n+1}^2 > 0.$$

(3) 计算得:

$$K = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & & & \\ & & \ddots & & \\ & & & \ddots & \\ & & & & 2 & -1 \\ & & & & -1 & 2 \end{pmatrix}$$

$$\text{令 } D = 2I_n \quad W = \begin{pmatrix} 0 & 1 & & & \\ & 1 & 0 & & \\ & & \ddots & \ddots & \\ & & & \ddots & 0 & 1 \\ & & & & 1 & 0 \end{pmatrix}$$

$$\text{则 } K = D - W$$

$$\text{设 } W\alpha = \lambda_0 \alpha \quad \alpha \neq 0 \quad \left(\begin{array}{l} W \text{ 对称} \\ \lambda_0 \text{ 实数} \end{array} \right)$$

$$\text{由圆盘定理. } |\lambda_0| < 2.$$

$$\Rightarrow K = 2(I_n - \frac{1}{2}W) \quad \rho(\frac{1}{2}W) < 1$$

(谱半径)

$$\begin{aligned} \Rightarrow K^{-1} &= \frac{1}{2} (I_n - \frac{1}{2}W)^{-1} \\ &= \frac{1}{2} (I_n + \frac{1}{2}W + \frac{1}{4}W^2 + \dots) \end{aligned}$$

$$\Rightarrow K^{-1} \text{ 是非负矩阵 (因为 } W \geq 0)$$

W 是图 $① \leftrightarrow ② \leftrightarrow ③ \leftrightarrow \dots \leftrightarrow ④$

的 adjacency 矩阵

W^k 的第 (i, j) 元素 ^{非 0} 意味从 $① \rightarrow ②$
的长度为 k 的路存在

因此 K^{-1} 是正矩阵.

注: 圆盘定理 $A = (a_{ij}) \in M_n(\mathbb{C})$
 λ 一个特征值, 则

$$|\lambda - a_{ii}| \leq \sum_{j \neq i} |a_{ij}| \quad i=1, \dots, n.$$

4. 证明:

$$\forall \varepsilon > 0, \text{ 令 } \tilde{A} = (P(A) + \varepsilon)^{-1} A$$

$$\text{则 } P(\tilde{A}) = \frac{P(A)}{P(A) + \varepsilon} < 1$$

$$\Rightarrow \lim_{k \rightarrow +\infty} \tilde{A}^k = 0$$

$$\exists N_1 > 0, \forall k \geq N_1, \|\tilde{A}^k\| < 1$$

$$\Rightarrow \|A^k\| < (P(A) + \varepsilon)^k$$

$$\Rightarrow \|A^k\|^{\frac{1}{k}} < P(A) + \varepsilon.$$

$$\text{令 } \bar{A} = (P(A) - \varepsilon)^{-1} A$$

$$\text{则 } P(\bar{A}) = \frac{P(A)}{P(A) - \varepsilon} > 1$$

$$\Rightarrow \|\bar{A}^k\| \rightarrow +\infty$$

$$\exists N_2 \in \mathbb{N}, \forall k > N_2,$$

$$\|\bar{A}^k\| > 1$$

$$\text{即 } \|A^k\| > (P(A) - \varepsilon)^k$$

$$\Rightarrow \|A^k\|^{\frac{1}{k}} > P(A) - \varepsilon$$

$$\text{令 } N = \max(N_1, N_2), \text{ 则}$$

$$\forall \varepsilon, \exists N, k \geq N \text{ 时}$$

$$P(A) - \varepsilon < \|A^k\|^{\frac{1}{k}} < P(A) + \varepsilon$$

注: $\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sigma_1$

(第一个奇异值或 $A^T A$ 最大特征值的平方根)

$$\Rightarrow \rho(A) \leq \sigma_1 = \|A\|$$

若 $\rho(A) > 1$, $\exists \alpha \quad A\alpha = \lambda\alpha$

$$|\lambda| = \rho(A) \quad \frac{\|A\alpha\|}{\|\alpha\|} = \rho(A)$$

$$\|A^k\| \geq \frac{\|A^k \alpha\|}{\|\alpha\|} = \rho(A)^k \rightarrow +\infty$$

若 $\rho(A) < 1$, $A^k \rightarrow 0 \quad (\text{当 } k \rightarrow +\infty)$

$$\Rightarrow (A^k)^T A^k \rightarrow 0 \Rightarrow \|A^k\| \rightarrow 0$$

即 $A_1^T A_2$ A_1^T 是 A 去掉
第 i 行, A_2 是 A 去掉第 j 列
 A_1^T 行满秩, A_2 列满秩.

$$\text{若 } A_1^T A_2 x = 0 \Rightarrow$$

5. 证明:

(1) 设 A 的特征值是
 $\lambda_1, \lambda_2, \dots, \lambda_n$ (包含重根)

则 A^k 特征值是

$$\lambda_1^k, \lambda_2^k, \dots, \lambda_n^k$$

因为 $A^k > 0$, 且 $\rho(A^k)$

$$= \rho(A)^k.$$

不妨设 $\lambda_1^k = \rho(A)^k \Rightarrow |\lambda_1| = \rho(A)$

存在 $\alpha_1 \neq 0 \in \mathbb{R}^n$ $A\alpha_1 = \lambda_1\alpha_1$

$$\Rightarrow A^k \alpha_1 = \lambda_1^k \alpha_1 = \rho(A^k) \alpha_1$$

但 $A^k > 0$, 由 Perron-Frobenius 定理, $\exists x > 0$, $A^k x = \rho(A^k) x$
且 $\alpha_1 = c_1 x$, $c_1 \in \mathbb{R}$, $\Rightarrow Ax = \rho(A)x$
 $\Rightarrow \lambda_1 = \rho(A)$.

由 $\lambda_1^k = \rho(A^k)$ 的代数重数 $= 1$,
则 $\lambda_1 = \rho(A)$ 的代数重数 $= 1$

(2) 设 $Ay \leq cy \quad y \geq 0$,

因为 A^T 有特征值 $\rho(A)$ 且 $A^T \geq 0$
 $(A^T)^k > 0$, 应用 (1), 存在 $w > 0$,

$$A^T w = \rho(A) w.$$

$$\text{即 } w^T A = \rho(A) w^T$$

$$\Rightarrow w^T A y = \rho(A) w^T y$$

$$\underline{\text{且}} \quad w^T A y \leq c w^T y$$

$$\Rightarrow \rho(A) \leq c$$

$$A y \leq c y \Rightarrow A^2 y \leq c A y \leq c^2 y$$

$$\Rightarrow A^k y \leq C^k y$$

因为 $A^k > 0$, $y \geq 0$,

$$\Rightarrow A^k y > 0$$

$$\Rightarrow C^k y > 0 \quad \text{因此 } y > 0.$$

6. (1) 因为 $\rho(A)$ 也是 A^T 的特征值,
且 $A^T > 0$, 由 Perron-Frobenius
定理, 存在 $w_0 > 0$,

$$A^T w_0 = \rho(A) w_0$$

设 $a = w_0^T v$, 则 $a > 0$ (因为
 $w_0, v > 0$) 令 $w = \frac{1}{a} w_0$

则 $w^T v = 1$, 且 $w^T A = \rho(A) w^T$.

$$(2) \text{ 令 } B = \frac{1}{\rho(A)} A$$

则 $\rho(B) = 1$, 且 $B > 0$

由 Perron-Frobenius 定理,

$\rho(B)=1$ 是 B 的特征值, 代数重数几何重数均为 1.

设 $P^{-1}BP = J = \begin{pmatrix} J_1 & & \\ & J_2 & \\ & & \ddots \\ & & & J_s \end{pmatrix}$

$J_1, \dots, J_s$ Jordan 块.

$$J_i = \begin{pmatrix} \lambda_i & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & \lambda_i \end{pmatrix} \quad i=1, \dots, s$$

不妨设 $\lambda_1=1$, 则 $J_1=(1)$

其余 $|\lambda_i| < 1 \quad i=2, \dots, s$.

$$\Rightarrow \lim_{t \rightarrow +\infty} B^t = \lim_{t \rightarrow +\infty} P \begin{pmatrix} 1 & & \\ & J_2^t & \\ & & \ddots \\ & & & J_s^t \end{pmatrix} P^{-1}$$

$$= P \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & 0 \end{array} \right) P^{-1} \quad (\text{因为})$$

$$\lim_{t \rightarrow +\infty} J_i^t = 0 \quad i=2, \dots, s)$$

$$\hat{=} B^\infty = \lim_{t \rightarrow +\infty} B^t.$$

$$\text{则 } \text{rank } B^\infty = 1. \quad \hat{=} B^\infty = \alpha \beta^T$$

$$\text{因为 } Bv = v, \text{ 则 } B^2v = Bv = v,$$

$$\Rightarrow B^\infty v = v \text{ 即 } \alpha \beta^T v = v.$$

$$\Rightarrow \alpha = c_1 v \exists c_1 \neq 0 \in \mathbb{R}.$$

$$\text{因为 } w^T A = P(A) w^T, \Rightarrow w^T B = w^T$$

$$\Rightarrow w^T B^\infty = w^T \text{ 即 } w^T (\alpha \beta^T) = w^T$$

$$\Rightarrow \beta^T = c_2 w^T \exists c_2 \neq 0 \in \mathbb{R}.$$

$$\text{即 } B^\infty = c_1 c_2 (v w^T)$$

$$\text{但 } w^T v = 1, w^T B^\infty = w^T$$

$$c_1 c_2 w^T v w^T = w^T \Rightarrow c_1 c_2 = 1$$

$$\text{因此 } B^\infty = v w^T$$

(3) $\frac{1}{5} \frac{7}{10}$

$$w^T A u = \lambda_0 w^T u$$

$\parallel$

$$P(A) w^T u$$

$$\Rightarrow (P(A) - \lambda_0) w^T u = 0 \Rightarrow w^T u \neq 0$$

($\ddot{+}$): $\lambda_0 \in \mathbb{C}$, $u \in \mathbb{C}^n$.

$$(4). \lim_{t \rightarrow +\infty} x_t = \left(\lim_{t \rightarrow +\infty} B^t \right) x_0$$

$$= (v w^T) x_0 = (w^T x_0) v.$$

7.

证明: 由 Perron-Frobenius 定理

存在 $\alpha > 0$, $A\alpha = \rho(A)\alpha$

因为 $B > A$, $B\alpha > \rho(A)\alpha$

$\exists \varepsilon > 0$ $B\alpha > (\rho(A) + \varepsilon)\alpha$

假设 $\rho(A) \geq \rho(B)$, 令

$$B^* = \frac{B}{\rho(A) + \varepsilon} \quad \text{则 } B^* > 0, \rho(B^*) < 1$$

且 $B^*\alpha > \alpha$

$$(B^*)^k \alpha > (B^*)^{k-1} \alpha > \cdots > \alpha > 0$$

当 $k \rightarrow +\infty$, $(B^*)^k \rightarrow 0$ (因为 $\rho(B^*) < 1$)
矛盾!

本题也可以使用 5 (2).

$$\text{设 } B\beta = P(B)\beta - \beta > 0$$

$$\text{则 } A\beta < B\beta = P(B)\beta$$

$$\text{令 } w > 0 \text{ 满足 } A^T w = P(A)w$$

$$w^T A \beta < P(B) w^T \beta$$

$$\parallel$$
$$P(A) w^T \beta$$

$$\text{因为 } w^T \beta > 0, \Rightarrow P(A) < P(B)$$

$$8. F(t+1) = L F(t)$$

$$\text{即 } f_1(t+1) = f_1(t)b_1 + f_2(t)b_2 + \cdots + f_n(t)b_n$$

$$f_k(t+1) = f_{k-1}(t)S_{k-1} \quad k=2, \dots, n$$

$$\textcircled{1} L \geq 0, \text{ 且存在 } m > 0, L^m > 0$$

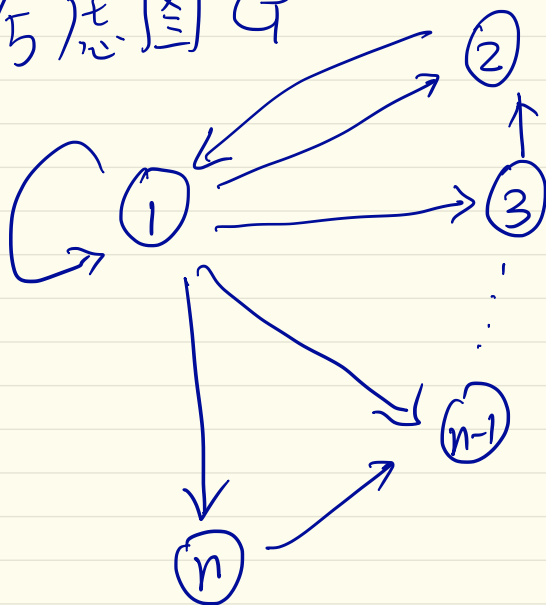
$$\text{设 } L = (l_{ij}) \quad L^k = (l_{ij}^{(k)})$$

$$l_{ij}^{(k)} = \sum_{h_1, \dots, h_{k-1}} l_{ih_1} l_{h_1 h_2} \cdots l_{h_{k-1} j} \geq 0$$

$$l_{ij}^{(k)} > 0 \Leftrightarrow \text{存在 } h_1, \dots, h_{k-1}$$

$$\text{使得 } l_{ih_1} > 0, l_{h_1 h_2} > 0, \dots, l_{h_{k-1} j} > 0$$

考虑图 G



$l_{st} > 0 \iff$ 存在一个边 $(s) \rightarrow (t)$

因此 $l_{ij}^{(k)} > 0 \iff$ 存在一条路

(路有方向): $(i) \rightarrow (j)$

② 由习题5, $\rho(L) \stackrel{=r}{=} \lambda$ 是 L 的特征值

$$\lim_{t \rightarrow +\infty} \left(\frac{L}{r} \right)^t = v w^T \quad (\text{习题6})$$

其中 $v, w > 0$

$$L v = r v, \quad L^T w = r w,$$

$$w^T v = 1. \quad (1, \dots, 1) v = 1$$

(分量之和 = 1)

$$f(t) = L^t f(0)$$

$$\lim_{t \rightarrow +\infty} \frac{f(t)}{r^t} = \left[\lim_{t \rightarrow +\infty} \left(\frac{L}{r} \right)^t \right] f(0)$$

$$= v w^T f(0) = (w^T f(0)) v$$

给定一个向量 $\alpha = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \geq 0$,

令 $|\alpha| = a_1 + \dots + a_n$.

$$F_k(t) = \frac{f_k(t)}{|f(t)|}$$

$$\lim_{t \rightarrow +\infty} F(t) = \lim_{t \rightarrow +\infty} \frac{f(t)}{|f(t)|}$$

$$= \lim_{t \rightarrow +\infty} \frac{\frac{f(t)}{r^t}}{\frac{|f(t)|}{r^t}} = \frac{(w^T f(0))v}{w^T f(0)}$$

$$= v$$

注: $w > 0, f(0) \geq 0 \Rightarrow w^T f(0) > 0$
但 $f(0) \neq 0$